

MATH 1342 – Elementary Statistics – Week 11: Inference for One Proportion and Two Proportions

Nov 2 – 4 | 6.1 Inference for a Proportion (Distribution, CI, Hypothesis Test) • 6.3 Inference for a Difference in Proportions (Distribution, CI, Hypothesis Test) | Fall 2026 ·
Section 019 · MW 7:30–8:50 PM · RRC Room 8210

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CO8) 2. Construct and interpret a confidence interval for a population proportion with z^* for 90%, 95%, or 99% confidence. (CO4, CO9) 3. Carry out a one-proportion z test and conclude in context, and explain why the CI standard error uses $\hat{p}$ while the test uses p_0. (CO4, CO10) 4. Classify a study as calling for one-proportion or difference-in-proportions inference, and check the conditions. (CO8) 5. Construct and interpret a confidence interval for $p_1 - p_2$, including what its sign means and whether it contains 0. (CO9) 6. Compute the pooled proportion and carry out a two-proportion z test from summary data or a two-way table. (CO10)	e-Text: 6.1-CI Confidence Interval for a Proportion	MO1, MO2, MO3	Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs	MO1, MO2, MO3	Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM)	MO1, MO2, MO3
		e-Text: 6.1-D Distribution of a Proportion	MO1, MO2, MO3				
		e-Text: 6.1-HT Hypothesis Test for a Proportion	MO1, MO2, MO3	Participation: Difference in Proportions – Try It: two-proportion randomization (18/30 vs. 12/30); treatment vs. placebo two-way table practice; compare simulation with the formula	MO4, MO5, MO6	Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 8, 11:59 PM)	MO4, MO5, MO6
		Interactive Lecture: 6.1 Inference for a Proportion	MO1, MO2, MO3				
		Slides: 6.1 Inference for a Proportion	MO1, MO2, MO3			Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)	None
		e-Text: 6.3-CI Confidence Interval for a Difference in Proportions	MO4, MO5, MO6				
		e-Text: 6.3-D Distribution of a Difference in Proportions	MO4, MO5, MO6				
		e-Text: 6.3-HT Hypothesis Test for a Difference in Proportions	MO4, MO5, MO6				
		Interactive Lecture: 6.3 Inference for a Difference in Proportions	MO4, MO5, MO6				
		Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization)	MO4, MO5, MO6				